Kutty Banerjee

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Summary:

Held several positions at NVIDIA across CPU, GPU, Chipset teams as an Architect, Individual Contributor and Chip Lead. Familiar with improving design at the SW/HW interface layer, leading Cross Functional features and familiar with the GPU driver stack.

Software Architect, NVIDIA [Dec 2010 ~ Present]

- **Tegra T132 SoC / Denver ARM CPU**
  - T132 is the Tegra SoC that replaces ARM A15 with NVIDIA's custom CPU code named Denver. **My role is to ensure we get the best Perf/Power out of this SoC on WoT and Tegra** especially for Browser / Graphics and Java workloads.
  - Denver uses a microkernel called MTS on the CPU and exposes a Threaded CPU to the OS. Designed scheduling changes in this microkernel to allow **Transparent Core Merging** of unused/underused Cores. This allows us to switch to the low power cluster more often on **Windows** which did not have native support for this at the time.
  - Investigated **Cluster Switching Perf latency** due to cache incoherence between clusters. Drove next gen HW changes to fix this.
  - Worked with SW/Architecture teams to implement a **QEMU based Memory Profiler**. Worked with Mozilla, Opera engineers to discuss findings about Web Workloads.
  - Implemented **Large Page Support** in Tegra Linux kernel to better Web Workload TLB coverage
  - **Optimized** Tegra kernel to not broadcast cpu cache snoop and TLB invalidates for browser **Thread Local Memory**. This betters Google V8 benchmark by 15% in addition to saving power
  - Scoped Tegra Driver/Android SW changes required to port to the upcoming ARM64 bit architecture.
- **Windows Tegra Driver / ARM port**
  - Tegra has a Unified CPU/GPU cache **incoherent framebuffer**. My role was to scope out and **coordinate Driver changes across OGL/DX/CUDA and GPU kernel** and make sure it was safe for this. Also made sure it was safe for ARM memory model.
  - Worked with HW L2/AXI teams to investigate cache flush/invalidate latencies and drive improvements for the nextgen SoC to **improve Graphics/Compute** Workload performance
  - Did scoping to understand impact of AXI Bus on GPU drivers especially GPU read/write ordering completions. Found a hw bug, implemented fixes throughout the Driver stack.
- **Project Orion**
  - NVIDIA's ARM CPU with GPU on the same SoC for HPC along with custom cache coherent interconnect. My role was to drive Uniform Virtual Address Space Implementation across Chipset, GPU HW and CUDA/GPU kernel teams.
  - Designed changes to AARCH64 Linux kernel to implement Shadow Page Tables

Senior System Software Engineer, NVIDIA [2005 June ~ Dec 2010]

- **GPU/Chipset Verification Lead**
  - The role was to reach out to the various HW unit leads, SW stake holders and make sure we had a coherent Chip Verification Strategy across HW/SW for future GPUs/Chipsets.
  - **PCIE Verification**: Drove changes to enable Directed Tests to run on Vista/XP for ASLM, ASPM, Standby, Hibernate, TDR, chipset deadlocks while reading CPU memory from GPU
  - **Power Mgmt Verification:** Designed a Probe Based Framework which allowed probes to be inserted into our Driver. Probes can introduce arbitrary waits/delays waiting for HW events. This allowed testing Power Transitions overlapped with interesting GPU states.

- **Display Bandwidth Verification**: Coordinated with Display HW leads and SW leads to allow testing Display scan out while stressful GPU activities occur concurrently prior to tapeout in RTL and Functional Model. This allowed avoiding Bandwidth Starvation Bugs.
- Increased coverage of trouble areas like **GPU context switching, Thread scheduling and ECC** using HW unit randoms and directed tests being fed real world instruction traces
- Put in place Code Coverage System for Driver code using Bullseye.

- **Fermi GPU Driver CHIP Lead**
  - Drove the GPU kernel team from POR Architecture through RTL/Emulation/Bringup and Release. This included leading a 15 member team and interfacing with other SW and HW teams.
- **Fermi GPU Driver Power Management**
  - Implemented P-state switching support. Worked with HW to get BLCG/ELCG support. Designed Engine Level Power Gating code, team of 2 implemented the change.
  - Helped scope out work for a team of 3 to offload code to a microcontroller to avoid CPU interrupt latency problems.
  - Worked with HW and SW unit leads to draft the first proposal for Vista Workload Throttling. This was later postponed until the next GPU generation.
- **Fermi GPU Driver Thread Scheduling Lead**
  - Designed and implemented 85% of the Scheduler code in the driver. This unit was redesigned from scratch in Fermi to avoid latency problems of the previous generation.
  - Implemented a Pipelined Scheduling Mode that allowed queued tasks to prefetch, TLB and cache warmup before the threads are executed. **Microsoft Vista DX10** feature.
  - Implemented separate **scheduling policies** to **saturate PCIE b/w for Graphics threads** and minimize Launch Latency for CUDA/PhysX/Video threads.
  - Designed the Error Recovery Subsysten and Power Aware Scheduling Subsystem. Mentored team members to implement the actual code (**Vista TDR** subsystem)
  - Implemented **Mid Buffer Preemption of Graphics work on Vista** for better User Experience
  - Debugged Perf and Latency bottlenecks
- **Fermi GPU Driver Kernel MMU / Memory Management**
  - Helped the MMU lead wire up memory management layer of the Fermi GPU kernel. This includes basic support for Multi level Page Table setup code, TLB invalidation logic, TLB invalidation batching optimizations, NOMMU mode support, fixed bugs.
- **Driver Non Stall Interrupts**
  - Implemented low latency lockless interrupt code path and ported all GPU notifiers to use this. This interrupt line is very rarely disabled and has low dependency on non ISR code.
- **Curie Architecture GPU Kernel Driver**
  - Debugged all parts of the GPU kernel, fixed bugs and triaged perf bottlenecks.

Education

MS Computer Science, Worcester Polytechnic Institute, MA [2003 Aug ~ 2005 May]

- **Thesis**: Remote Execution for 3D Graphics. A distributed 3D Rendering pipeline (RMESA) to offload floating point intensive operations to a cluster to save Power on Mobile Devices
- Recipient of **WPI Institute Fellowship Award** for outstanding research work.

BS Computer Science, University of Mumbai **[1999 July ~ 2003 May]**

Publications

- **PowerSpy: Fine-Grained Software Energy Profiling for Mobile Devices**
  Kutty Banerjee, Emmanuel Agu
  International Conference on Wireless Networks, Communications and Mobile Computing, 2005
- **A Middleware Architecture for Mobile 3D Graphics**
  Emmanuel Agu, Kutty Banerjee, Shirish Nilekar, Oleg Rekutin, Diane Kramer
  International Journal of Parallel, Emergent and Distributed Systems Volume 21, Issue 3, 2006
- **Estimating Mobile Memory Requirements and Rendering Time for Remote Execution of the Graphics Pipeline**
  Kutty Banerjee, Fan Wu, Emmanuel Agu - Proceedings of Eurographics 2005
- **Remote Execution for 3D Graphics on Mobile Devices**
  Kutty Banerjee, Emmanuel Agu
  International Conference on Wireless Networks, Communications and Mobile Computing, 2005
- **A Middleware Architecture for Mobile 3D Graphics**
  Emmanuel Agu, Kutty Banerjee, Shirish Nilekar, Oleg Rekutin, Diane Kramer
  Third International Workshop on Mobile Distributed Computing, 2006

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